

12 (**short**) In the C++ code below, `f1` and `f2` are both designed to calculate a^n , where a is a real number and n is an integer. There is also a main program which calls `f1` and `f2`.

```
float f1(float a, int n) {
    float b = 1.0;
    while (n>0) {
        b = b*a;
        n = n-1;
    }
    return b;
}

float f2(float a, int n) {
    return exp(n*log(a));
}

int main() {
    cout << f2(7.0, 2) - f1(7.0, 2) << endl;
    return 0;
}
```

- (a) What is the algorithmic complexity of `f1` in terms of n ? What is the algorithmic complexity of `f2` in terms of n ? [4]
- (b) What, approximately, would you expect the main program to print, and why? [3]
- (c) For what values of a and n will `f1` and `f2` fail catastrophically? [3]

END OF PAPER

Part IA 2016
Paper 4: Mathematical Methods
Numerical Answers

Section A:

1(a) $-1/3$; (b) $-\pi/8$

3 volume is 56

4(c) Eigenvalues are $-a, -a, 2a$ Eigenvector for the largest eigenvalue is $(1/\sqrt{3}, 1/\sqrt{3}, 1/\sqrt{3})$

Section B:

9 (0, 3) – Maximum; (0, -3) – Minimum,

$(12/\sqrt{41}, 15/\sqrt{41})$ and $(-12/\sqrt{41}, -15/\sqrt{41})$ are saddle points

Section C:

12(b) the main program will display 7.62939e-06